

What Makes a Video Game a “Top-Seller”

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The global video game business is a multi-billion dollar industry and has been for many years. As technology improves, so does gameplay, which in turn, encourages further profit potential. In 2024 the revenue from the global gaming market was estimated at almost 455 billion U.S. dollars (Statista, n.d.). Despite the large revenue and profit potential of video games in general, a number of game releases fail to achieve commercial success. Game developers and publishers would benefit from the ability to predict which games will perform well before launch, and greatly improve decisions regarding investments, marketing, reducing risk, and maximizing profitability. The goal of this analysis is to investigate whether we could build a predictive model using global sales data to identify top-selling games.

Data

The dataset, hosted on Kaggle, aggregates video game sales and ratings based on Metacritic scores and provides insight into global video game sales and ratings. Initially, the dataset contained 16,719 entries. After cleaning and removing rows with missing values, we retained 6,825 observations. The decision to not impute missing values was made primarily because this dataset was comprised of two datasets that were merged. Imputation may have introduced additional bias, and the goal was to focus on applying machine learning models. The focus was primarily on features available prior to release: platform, year of release, genre, publisher, developer, critic and user scores, and rating. The outcome variable `Top_Seller_75` was engineered to classify games that ranked in the top 25% of global sales.

Methods

We utilized two supervised machine learning models, logistic regression and random forest. The dataset was split into training and testing sets (80/20). To ensure the reliability and generalizability of our results, we implemented a 5-fold cross-validation strategy during model training. The analysis

pipeline began with comprehensive feature encoding and selection to prepare the dataset and highlight the most informative predictors. Following this, we trained and tested multiple classification models, evaluating their performance using a range of key metrics, including accuracy, precision, recall, and F1 score. In addition to these standard metrics, we conducted a ROC-AUC curve analysis to assess the models' ability to distinguish between classes across various threshold settings. To further examine model performance, we reviewed the confusion matrix, which provided insight into the distribution of true positives, false positives, true negatives, and false negatives. Finally, to optimize model performance, we conducted hyperparameter tuning using GridSearchCV, allowing us to systematically explore and identify the best combination of parameters for each model. All data cleaning, analysis, and modeling were performed using Python, utilizing libraries such as pandas for data manipulation, seaborn and matplotlib for visualization, and scikit-learn for machine learning algorithms and evaluation metrics.

Results Discussion

Exploratory data analysis showed extensive variation in global video game sales. On average, games in the processed dataset sold about 0.78 million units, but the distribution was heavily right skewed. Most games had fairly low sales, while a small number sold tens of millions, with a few outliers even surpassing 80 million units. The outliers were not removed when judged to be valid blockbuster hits that dominated the global market. Figure 1 illustrates the distribution of global sales, limited to games that sold fewer than 5 million units, to better highlight the concentration of lower-selling titles without the distortion caused by extreme outliers.

When we looked at which genres had the highest rate of top-seller performance, Misc, Shooter, and Sports stood out. Approximately 32% of Misc games, 30% of Shooter games, and 28% of Sports games in the dataset qualified as top sellers. It is worth noting that Action games were by far the most common genre overall, with 1,630 titles in the dataset, followed by Sports and Shooter games. However,

having the most titles does not mean having the highest rate of commercial success. This pattern is captured in Figure 2, which shows the proportion of top sellers within the pool of top-performing titles by genre, offering a clearer view of genre-based performance trends in the market.

I started with a basic logistic regression model and used cross-validation to evaluate how well it could classify top sellers. The model showed stronger recall than precision. Specifically, precision for identifying top sellers was approximately 0.50, while recall reached approximately 0.70, meaning the model was reasonably good at catching actual top sellers but also produced a notable number of false positives. The 5-fold cross-validation confirmed these results, with an average precision of 0.474 and average recall of 0.703. Figure 3 presents the ROC curve for the logistic regression model, showing an AUC of 0.80, which indicates solid discriminatory power, though not optimal.

When I switched to a random forest classifier and tuned the hyperparameters using GridSearchCV, performance improved substantially. The best parameters identified were a max depth of 10 and 300 estimators. These tree-based models were better at capturing more complex patterns in the data and interactions between variables. For the tuned model, precision and recall for top sellers both reached 0.69, producing an F1 score of 0.69 for that class and an overall weighted F1 of 0.84. Looking at the confusion matrix for the tuned random forest model (Figure 4), we can see that most of the model's misclassifications occurred in borderline cases, meaning games that just barely made or missed the threshold for being considered a top seller.

Genre and platform were among the most commonly assumed predictors of sales success, but the random forest model revealed that user and critic engagement metrics were actually more influential. As shown in the feature importance plot (Figure 5), User_Count, Critic_Score, and Critic_Count ranked highest, outweighing genre, publisher, and even platform in predictive power. This finding suggests that how a game is received, even before or shortly after release, is more telling of its commercial success than its category or release platform.

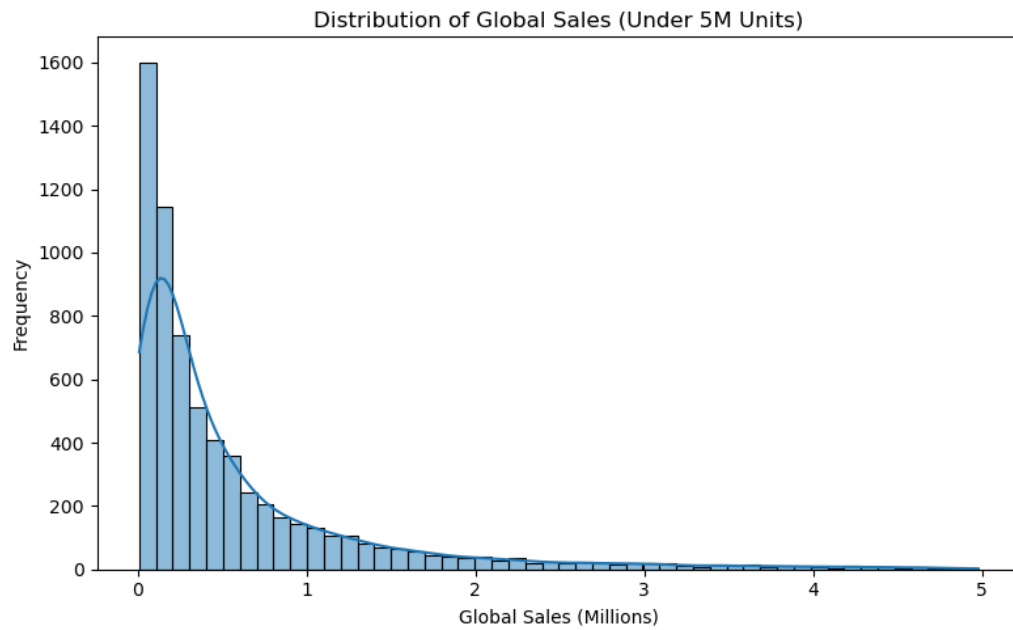
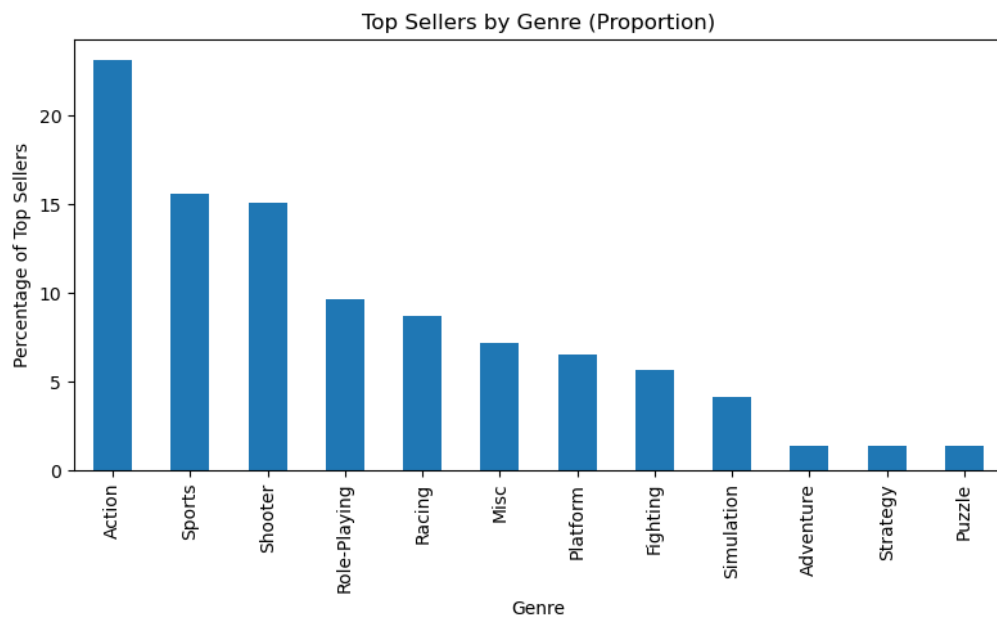
Figure 1 Distribution of Global Sales (Games Under 5 Million Units)*Figure 2 Top Sellers by Genre (Proportion)*

Figure 3 ROC Curve for Logistic Regression

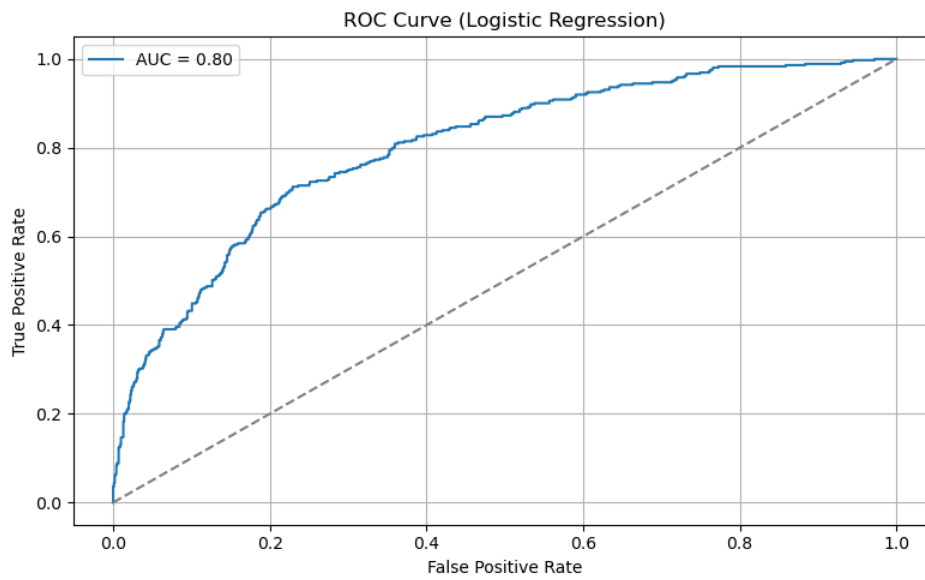


Figure 4 Confusion Matrix (Tuned Random Forest Classifier)

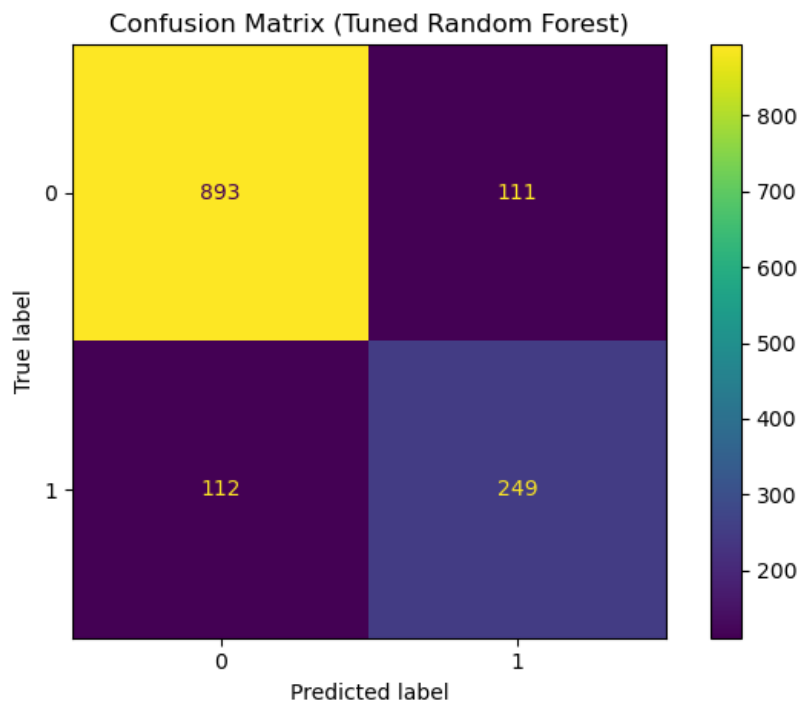
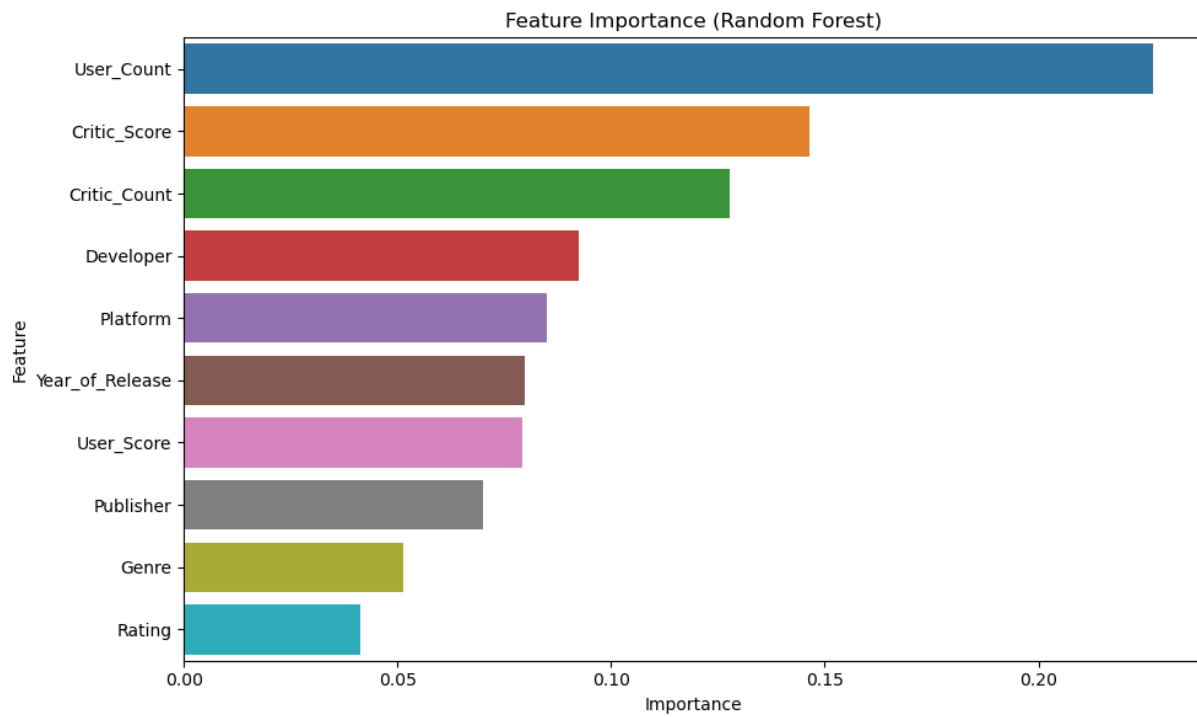


Figure 5 Feature Importance from Random Forest Model

Strengths, Limitations, and Future Work

A key strength of this project was the use of real-world publicly available data from two merged datasets, which offered a diverse range of games and features for analysis. The application of multiple machine learning models and the use of cross-validation and grid search helped ensure robust performance comparisons.

However, the project also faced limitations. The exclusion of rows with missing values significantly reduced the dataset size, which may have impacted model generalizability. Additionally, the dataset lacked important predictors like marketing budgets, release timing, and user reviews, which are factors that can strongly influence game sales. Another key limitation was the historical nature of this dataset. The video game industry is constantly evolving, and with trends shifting in the industry, model performance may drop over time.

For future work, incorporating additional data sources, such as social media sentiment or review metadata, could enhance prediction accuracy. Experimenting with advanced models like gradient boosting or neural networks, and performing time-based splits to mimic real forecasting conditions, would also be valuable next steps.

Conclusion

This project helped uncover what makes a video game a top seller, and the answer is not always simple. Sales were heavily skewed, with most games selling modestly while a few blockbusters dominated the market. Genre made a difference in terms of hit rate. Misc, Shooter, and Sports titles showed the highest proportion of top sellers within their respective genres, while Action games dominated in sheer volume.

I started with logistic regression, which established a useful baseline. The model achieved solid recall around 0.70, meaning it was reasonably good at identifying true top sellers, but its precision of approximately 0.50 indicated it also flagged many false positives. Switching to a random forest model

and tuning it using GridSearchCV improved performance significantly. The tuned model achieved an F1 score of 0.69 for top sellers and a weighted F1 of 0.84, with both precision and recall for the top seller class reaching 0.69. The tuned model still struggled in borderline cases, but overall it performed more reliably and consistently than the logistic regression baseline.

What really stood out from the feature importance plot was that user and critic engagement were strong indicators of success. User_Count and Critic_Score were more important than even genre or publisher, which shows how much buzz and reception matter in the gaming world.

This whole process reinforced how valuable it is to explore the data deeply, test different models, and experiment with tuning. It was not just about getting a high accuracy score. It was about understanding what drives game success and how those insights can help future decisions in game development or marketing.

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